ApproxDA-TransUNet: Exploring Context-Dependent Attention Approximation in Medical Image Segmentation

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***Abstract*—We introduce Global Context Requirement (GCR), a task-level property capturing the degree to which accurate segmentation depends on long-range spatial dependencies, and propose that GCR governs the sensitivity of performance to attention window size: high-GCR tasks require careful window tuning to achieve gains; low-GCR tasks are window-robust and benefit from approximation broadly. To validate this hypothesis, we design ApproxDA-TransUNet, a practical efficient dual-attention architecture built around three orthogonal approximation operators—windowed PAM (spatial scope M), low-rank projection (representation fidelity r), and grouped CAM (channel scope G)—reducing per-GPU training memory from 11.5 GB to 6.4 GB and enabling multi-GPU distributed training. Experiments on three benchmarks confirm the GCR hypothesis: with task-appropriate window selection, the same approximation framework improves both multi-organ CT segmentation (+1.14% vs. DA-TransUNet on Synapse with M=28; 80.94% DSC, high GCR) and polyp segmentation (+1.73% on Kvasir-SEG with M=56; 90.17% DSC, low GCR). Crucially, performance sensitivity to window size is 3.6× higher on the high-GCR benchmark (DSC range 2.30% vs. 0.64%), confirming GCR as a predictor of window-tuning importance rather than approximation direction. A secondary finding reveals gradient symmetry collapse: at initialization $g \approx 0.5$, symmetric branch gradients drive convergent representations and near-zero gate updates, pinning $g \approx 0.5$ regardless of window size, confirmed independently at both M=7 and M=14. These results shift the research question from *how to approximate* to *when to approximate*, with GCR providing a practical task-level criterion.**

***Index Terms*—medical image segmentation, dual attention, efficient transformers, attention approximation, gate collapse, global context**

I. Introduction

Transformer-based architectures have advanced medical image segmentation by capturing long-range spatial dependencies through dual attention—jointly modeling positional and channel correlations [1]. Methods such as DA-TransUNet [2] integrate these into hybrid CNN-Transformer backbones, achieving strong performance across multiple benchmarks. However, dual attention carries quadratic computational complexity, and a large body of efficient approximations—windowed attention [3], low-rank factorization [4], grouped channel interaction [5]—has emerged as a standard design response. These strategies share a common assumption: that approximation is universally applicable, with at most minor accuracy trade-offs. For medical image segmentation, which spans tasks with fundamentally different spatial reasoning requirements, this assumption has rarely been examined:

> *When is attention approximation safe for medical image segmentation?*

To investigate this systematically, we propose **ApproxDA-TransUNet**, a dual-attention architecture with three orthogonal approximation operators—windowed PAM, low-rank projection, and grouped CAM—each independently controllable. Rather than optimizing for peak performance, the design serves as a *scientific instrument*: any single approximation axis can be varied while the others remain fixed, enabling isolation of each operator's contribution.

Experiments reveal a consistent and task-dependent pattern. With appropriate window selection, ApproxDA improves over the full dual-attention baseline on both benchmarks: +1.14% DSC on multi-organ CT (Synapse, M=28) and +1.73% on polyp segmentation (Kvasir-SEG, M=56). Crucially, performance sensitivity to window size differs 3.6× across tasks—Synapse DSC spans 2.30% across window values while Kvasir spans only 0.64%—indicating that high-GCR tasks require careful window tuning while low-GCR tasks are window-robust. We additionally find that learnable routing mechanisms for weighting attention branches consistently collapse to a fixed equilibrium due to gradient symmetry—a fundamental property of symmetric two-branch architectures, not a training artifact.

Motivated by these observations, we hypothesize that a latent task property—which we term **Global Context Requirement (GCR)**—governs *window sensitivity*: high-GCR tasks exhibit steep DSC-vs-window curves and require careful tuning; low-GCR tasks are near-flat and window-robust. We formalize this hypothesis precisely in Section V.

The main contributions of this paper are:

- We design **ApproxDA-TransUNet**, a controllable dual-attention architecture with three independently adjustable approximation operators—windowed PAM, low-

rank projection, and grouped CAM—each independently controllable to isolate individual approximation effects. The design serves as a scientific instrument for studying when approximation is appropriate across tasks with different contextual requirements.
- We introduce **Global Context Requirement (GCR)** as a task-level concept and show that it governs window sensitivity: high-GCR tasks exhibit 3.6× higher sensitivity to window size than low-GCR tasks (range 2.30% vs. 0.64%), providing evidence for GCR as an explanatory factor governing how carefully the attention window must be tuned for each task.
- We derive the gradient dynamics of symmetric dual-branch gating and empirically verify **gradient symmetry collapse**: the routing gate converges to a near-constant value regardless of window size or rank—a fundamental limitation of symmetric two-branch attention design with broad implications for routing mechanism design.

The remainder of this paper is organized as follows: Section II reviews related work; Section III presents the ApproxDA framework; Section IV reports experimental results; Section V provides discussion and analysis; Section VI concludes.

II. Related Work

A. CNN-based Medical Image Segmentation

Convolutional encoder-decoder architectures have established the foundational design paradigm for medical image segmentation. U-Net [6] introduced a symmetric encoder-decoder with skip connections that combine deep semantic features with shallow spatial detail, enabling precise boundary localization. U-Net++ [7] densified the skip pathways through nested sub-networks to reduce the semantic gap between encoder and decoder. Attention U-Net [8] integrated soft attention gates on skip connections to selectively suppress irrelevant activations at each scale. ResUNet [9] incorporated residual connections for stable training at greater depth, while MultiResUNet [10] replaced standard convolutions with multi-resolution blocks for richer local feature representations. These five architectures form the primary CNN baselines evaluated in this work. Despite their strong performance on local-feature-dominant tasks, their limited receptive fields hinder long-range anatomical reasoning, motivating the Transformer-based methods discussed next.

B. Transformer-based Medical Image Segmentation

Transformer architectures have significantly advanced medical image segmentation by enabling global dependency modeling beyond the locality of convolutional neural networks. Representative methods such as TransUNet [11], Swin-Unet [12], UNETR [13], UCTransNet [14], and DAEformer [15] combine convolutional feature extraction with transformer-based contextual modeling to achieve state-of-the-art segmentation performance across a wide range of medical imaging tasks.

C. Dual Attention, DA-TransUNet, and Adaptive Routing

Position Attention Modules (PAM) and Channel Attention Modules (CAM) were introduced by DANet [1] for scene segmentation. PAM computes a non-local affinity matrix over all $N{=}H{\times}W$ spatial positions at $\mathcal{O}(N^2C)$ complexity, enabling each location to aggregate context from every other position. CAM models dependencies among all C channels through a channel-wise affinity matrix at $\mathcal{O}(C^2N)$ complexity, capturing global semantic channel correlations independent of spatial layout. Outputs of both branches are fused via learned coefficients to produce the final dual-attended representation.

DA-TransUNet [2] adapted this design for medical image segmentation by inserting dual-attention blocks at multiple skip connections and the bottleneck of a hybrid R50-ViT-B/16 encoder-decoder, with a learnable channel-wise gate fusing PAM and CAM at each block. DA-TransUNet achieves state-of-the-art accuracy on multiple benchmarks, but inherits the quadratic complexity of PAM and CAM, and its public implementation contains a DataParallel incompatibility that prevents multi-GPU training.

Learnable gating for feature branch fusion appears widely in transformer-based segmentation [5], [15], with the implicit assumption that such gates converge to meaningful, sample-adaptive strategies after end-to-end optimization. Existing work generally does not investigate whether the routing module remains effective or degenerates. This work explicitly analyzes the optimization dynamics of the gate within DA-TransUNet and shows that gradient symmetry between the two attention branches causes the gate to collapse to a near-constant coefficient, degenerating from adaptive routing into a fixed blend.

While existing work extensively investigates how to approximate attention more efficiently, a complementary question remains largely unexplored: *under what conditions is attention approximation fundamentally appropriate?* Prior efficient attention studies optimize for computational savings with the implicit assumption that approximation is universally safe, and learnable routing mechanisms are adopted without examining whether they remain effective after optimization. This work addresses both gaps by reformulating the dual-attention module of DA-TransUNet [2] into a controllable approximation framework, analyzing when approximation helps or hurts across tasks with different contextual requirements, and exposing the optimization dynamics that cause adaptive routing to degenerate into fixed blending.

III. Method

This section introduces the overall architecture of ApproxDA, then describes the three controllable approximation operators used to reformulate the dual-attention module, and finally analyzes their computational complexity.

A. Architecture Overview

ApproxDA-TransUNet adopts the hybrid CNN–ViT encoder-decoder structure of DA-TransUNet [2], with an R50-ViT-B/16 backbone pretrained on ImageNet-21K. Three

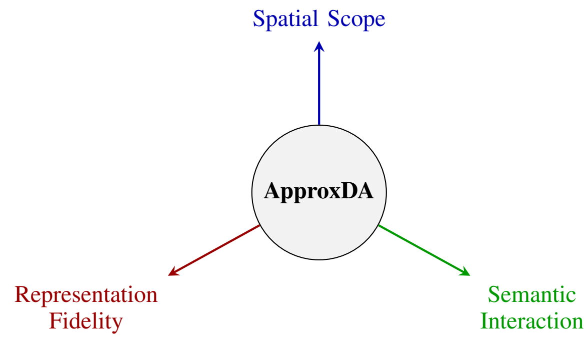

Fig. 1. ApproxDA-TransUNet architecture. ApproxDABlocks (Low-Rank Windowed PAM + Grouped CAM + gate routing) replace dual-attention modules at all skip connections and the bottleneck. `gate_mode` controls the routing during ablation.

Fig. 2. ApproxDA's three-dimensional approximation space. Each axis is independently controllable. Spatial scope (M) most directly governs long-range context capture and thereby approximation effectiveness on high-GCR tasks.

CNN encoder stages produce multi-scale features at strides 4, 8, and 16; a 12-layer ViT processes 14×14 patch tokens. ApproxDABlocks replace all dual-attention modules at three skip connections—512 channels at 28×28, 256 at 56×56, and 64 at 112×112—and at the bottleneck.

ApproxDA-TransUNet is designed around an explicit hypothesis: that three orthogonal approximation operators—windowed PAM, low-rank projection, and grouped CAM—can replace the quadratic DA module to yield both a practical efficient architecture and a systematic framework for studying when approximation is appropriate. Each operator targets a distinct approximation axis (spatial scope, representation fidelity, channel scope), so any two can be held fixed while the third varies. This makes ApproxDA not merely a faster model, but a model whose design enables controlled ablation of approximation effects across tasks with different contextual requirements.

Figure 1 illustrates the overall ApproxDA architecture.

B. Attention Approximation Formulation

The original DA module performs dense spatial and channel attention over the full feature map, with quadratic complexity in both spatial resolution and channel dimension. ApproxDA reduces this cost along three independent axes, each targeting a distinct aspect of the approximation:

Spatial scope (window size M) controls *how far* PAM reaches. Full global attention ($M{=}H$) captures all long-range spatial relationships; restricting to $M\times M$ local windows trades long-range context for lower complexity. This axis most directly governs whether high-GCR tasks retain the anatomical context they require.

Representation fidelity (rank r) controls *how precisely* attention is computed within each window. Low-rank projection ($r \ll M^2$) approximates the affinity matrix at reduced cost; higher r approaches exact attention.

Semantic interaction (groups G) controls *which channels* may interact in CAM. Partitioning C channels into G independent groups reduces cross-channel dependency while retaining within-group semantic interaction.

These axes correspond to three distinct types of approximation: *spatial approximation* (how far attention reaches, controlled by M), *representation approximation* (how precisely affinity is computed, controlled by r), and *channel approximation* (how broadly channels interact, controlled by G). The axes are fully independent: any two of $\{M, r, G\}$ can be held fixed while the third varies, enabling clean ablations that isolate individual approximation effects. Figure 2 visualizes this three-dimensional approximation space.

C. Spatial Approximation

Instead of computing global position attention, ApproxDA partitions the feature map into non-overlapping windows of size $M\times M$. Attention is computed independently inside each local window, reducing complexity from $\mathcal{O}(N^2C)$ to $\mathcal{O}(NM^2C)$.

Window partitioning. Following [3], we partition the feature map into n_W non-overlapping $M\times M$ windows. Window clamping $M \leftarrow \min(M, H, W)$ allows $M{=}112$ to yield global attention at all scales, enabling a clean ablation that isolates windowing from low-rank as the performance bottleneck.

Low-rank projection. Following [4], query and key features inside each window are projected from $N_w{=}M^2$ to rank $r \ll M^2$ via a learned matrix $\mathbf{W}_r \in \mathbb{R}^{r\times N_w}$:

$$\tilde{\mathbf{C}} = \mathbf{W}_r\mathbf{C}^\top, \quad \tilde{\mathbf{D}} = \mathbf{W}_r\mathbf{D}^\top \in \mathbb{R}^{r\times C}. \tag{1}$$

The attention matrix $QK^\top \in \mathbb{R}^{N_w\times N_w}$ is approximated by $Q_rK_r^\top \in \mathbb{R}^{N_w\times r}$, where $Q_r, K_r \in \mathbb{R}^{N_w\times r}$. Total complexity is $\mathcal{O}(NrC)$. Combined, the two approximations reduce attention storage at 112×112 from $\approx$15 GB to $\approx$38 MB ($\sim$390$\times$).

These two hyperparameters—window size M and projection rank r—jointly constitute the spatial approximation operator and form the primary experimental axes investigated on Synapse.

D. Channel Approximation

The original Channel Attention Module computes a $C\times C$ channel attention matrix at $\mathcal{O}(C^2N)$ cost. ApproxDA approximates this by partitioning C channels into G groups of width $C_g = C/G$, computing independent per-group attention:

$$X_{ji}^{(g)} = \frac{\exp(\mathbf{A}_i^{(g)}\cdot\mathbf{A}_j^{(g)})}{\sum_{k=1}^{C_g}\exp(\mathbf{A}_k^{(g)}\cdot\mathbf{A}_j^{(g)})}, \tag{2}$$

with total cost $\mathcal{O}(C^2N/G)$. Compared with dense channel interaction, grouped attention significantly reduces computational complexity while preserving local channel dependency.

The grouping number G provides another controllable approximation dimension, allowing the influence of channel approximation to be studied independently from spatial approximation.

E. Routing Strategy

The learnable routing mechanism of DA-TransUNet [2] is retained and extended to a configurable `gate_mode`. The outputs of the spatial and channel attention branches are fused via:

$$\mathbf{y} = \mathbf{W}_f\Big(g \cdot \hat{\mathbf{P}} + (1-g)\cdot \hat{\mathbf{C}}\Big) + \mathbf{x}, \tag{3}$$

where $\hat{\mathbf{P}}$ and $\hat{\mathbf{C}}$ are the normalized PAM and CAM outputs, $\mathbf{W}_f$ is a $1{\times}1$ convolution, and g is determined by the `gate_mode` parameter:

`pam`:	$g = 1.0$	(PAM only)
`cam`:	$g = 0.0$	(CAM only)
`fixed`:	$g = 0.5$	(equal blend)
`learn`:	$g = \sigma(\mathbf{W}_g \cdot \mathrm{GAP}(\mathbf{x})) \in (0,1)^C$	

This single-parameter design enables systematic ablation of routing strategy without any other architectural change. ApproxDA does not treat this routing module as a primary architectural contribution; instead, it serves as an analyzable component whose optimization behavior is investigated in Section V. As demonstrated later, the routing coefficient consistently converges to an almost constant value, revealing an unexpected optimization phenomenon.

F. Loss Function

The training objective combines Dice loss and cross-entropy loss with equal weights:

$$\mathcal{L} = \tfrac{1}{2}\mathcal{L}_{\text{Dice}} + \tfrac{1}{2}\mathcal{L}_{\text{CE}}. \tag{4}$$

These approximations reduce layer-level attention complexity: windowed PAM from $\mathcal{O}(N^2C)$ to $\mathcal{O}(NrC)$ ($r \ll N$); grouped CAM from $\mathcal{O}(C^2N)$ to $\mathcal{O}(C^2N/G)$. Since backbone operations dominate end-to-end computation, the primary value of ApproxDA is not whole-network FLOPs reduction but a controllable formulation for systematically isolating each approximation operator's effect on segmentation performance (Section IV).

IV. Experiments

We evaluate ApproxDA-TransUNet on three medical image segmentation benchmarks spanning different global context requirements. The experiments systematically examine computational efficiency, task-dependent approximation behavior across GCR regimes, the dynamics of symmetric gate routing, and the relationship between approximation strength and task context.

TABLE I
Dataset Properties and Global Context Requirement (GCR)

Dataset	Classes	Object Type	Long-range Context	GCR
Synapse	8	Multi-organ CT	High (inter-organ)	High
Kvasir-SEG	1	Polyp	Low (local boundary)	Low
ISIC 2018	1	Skin lesion	Low (texture/edge)	Low

High-GCR tasks require cross-organ anatomical reasoning; low-GCR tasks are dominated by local appearance cues. GCR governs approximation effectiveness (Section V).

A. Experimental Setup

Datasets. We evaluate ApproxDA on three representative medical image segmentation benchmarks covering different contextual characteristics.

Synapse Multi-Organ CT [16]: 30 abdominal CT scans with an 18/12 train/test split covering 8 organ classes (9 classes with background). Evaluation metrics: mean DSC and HD95.

Kvasir-SEG [17]: 1000 endoscopy polyp images (binary segmentation) with an 800/200 train/test split. Evaluation metrics: DSC and mIoU.

ISIC 2018 [18]: 2594 dermoscopy skin lesion images (binary segmentation) with a 2075/519 train/test split (80/20). Evaluation metrics: DSC and mIoU.

Table I summarizes the three datasets alongside their contextual characteristics and GCR classification, which motivates the cross-task analysis in Section V.

Implementation Details. All experiments use an R50-ViT-B/16 backbone pretrained on ImageNet-21K. Training uses SGD (momentum 0.9, weight decay 10^{-4}), initial learning rate 0.01 with polynomial decay, batch size 24, image size $224{\times}224$, 300 epochs, and seed 1234. The DA-TransUNet baseline runs on T4×1. ApproxDA-TransUNet runs on T4×2 via `torchrun` DDP with per-GPU batch 12 and NCCL all-reduce, under identical total batch size and base learning rate. Default ApproxDA configuration: $M{=}7$, $r{=}32$, $G{=}8$. Our DA-TransUNet re-run achieves validation DSC 0.795, consistent with the paper-reported 0.798, confirming faithful replication.

B. Computational Efficiency of ApproxDA

Table II compares the computational characteristics of ApproxDA and DA-TransUNet on Synapse. Although per-GPU VRAM is reduced substantially—from 11.5 GB to 6.4 GB—total GFLOPs are slightly higher (32.1 vs. 30.2) because the shared ViT backbone dominates total computation and Conv1d/Linear projection overhead partially offsets decoder-level savings. The reduced memory footprint enables multi-GPU distributed training via DDP, which DA-TransUNet does not support.

C. Task-dependent Behavior of Attention Approximation

We examine whether efficient attention approximation behaves consistently across medical segmentation tasks with different contextual requirements. Tables III–V compare

TABLE II
EFFICIENCY COMPARISON ON SYNAPSE (T4). GFLOPS MEASURED BY `FVCORE` (SINGLE 224×224 SLICE, INCLUDES ATTENTION `BMM`).

Method	Params (M)	GFLOPs	Train VRAM	Train Time	Infer (s/vol)	Multi-GPU
DA-TransUNet [2]	107.95	30.2	11.5 GB	11.4h ×1	120.0	No†
ApproxDA (gate=pam, M=7, r=32)	112.98	32.1	6.4 GB/GPU	8.3h ×2	121.3	Yes (DDP)
ApproxDA (gate=learn, M=7, r=32)	114.90	32.1	10.6 GB	11.5h ×1	114.2	Yes (DDP)

†DA-TransUNet does not support multi-GPU training. ApproxDA uses DDP (`torchrun`, NCCL); per-GPU batch 12 (total 24), LR unchanged.

TABLE III
SEGMENTATION RESULTS ON SYNAPSE MULTI-ORGAN CT

Method	DSC (%) ↑	HD95 (mm) ↓
CNN-based		
U-Net [6]	76.85	39.70
U-Net++ [7]	76.91	36.93
Residual U-Net [9]	76.95	38.44
Att-Unet [8]	77.77	36.02
MultiResUNet [10]	77.42	36.84
Transformer-based		
TransUNet [11]	77.48	31.69
UCTransNet [14]	78.23	26.75
TransNorm [19]	78.40	30.25
MIM [20]	78.59	<u>26.59</u>
Swin-Unet [12]	79.13	21.55
DA-TransUNet [2]	**79.80**	**23.48**
Dual-attention approximation (ours)		
ApproxDA-TransUNet	<u>80.94</u>	<u>27.49</u>

Baselines from [2] Table 1 (SGD, 500ep, 224×224). ApproxDA: gate=pam, M=28, r=32, G=8, SGD 300ep (window sensitivity ablation identifies M=28 as optimal). We also re-ran DA-TransUNet under our protocol (SGD, 300ep) and obtained 72.03% DSC; the gap from paper-reported (79.80%) is primarily due to optimizer and epoch differences (Adam/500ep vs. SGD/300ep), so we report paper-reported values for all Synapse baselines.

TABLE IV
SEGMENTATION RESULTS ON KVASIR-SEG POLYP

Method	DSC (%) ↑	mIoU (%) ↑	HD95 (mm) ↓
U-Net [6]	88.22	80.12	–
Att-Unet [8]	86.61	78.01	–
U-Net++ [7]	86.57	77.67	–
Res-UNet [9]	77.85	66.04	–
TransUNet [11]	87.91	80.03	–
DA-TransUNet [2]†	<u>88.44</u>	<u>81.70</u>	<u>53.04</u>
ApproxDA-TransUNet (ours)	**90.17**	**84.27**	**44.35**

Other baselines from [2] Table 2 (Adam, 500ep, 75/25 split); HD95 not reported. †DA-TransUNet re-run under the same protocol as ApproxDA (SGD, 300ep, 80/20 split). Paper-reported [2]: 88.47% DSC, 81.02% mIoU. ApproxDA: gate=pam, M=56, r=32.

TABLE V
SEGMENTATION RESULTS ON ISIC 2018 SKIN LESION

Method	DSC (%) ↑	mIoU (%) ↑
U-Net [6]	87.22	81.14
Att-Unet [8]	87.60	81.51
U-Net++ [7]	87.30	81.33
Res-UNet [9]	83.32	76.51
TransUNet [11]	<u>88.78</u>	<u>82.63</u>
DA-TransUNet [2]	**88.88**	**82.78**
ApproxDA-TransUNet (ours)	*pend.*	*pend.*

Baselines from [2] Table 2 (Adam, 50ep, 75/25 split). ApproxDA: SGD, 300ep, 80/20 split.

ApproxDA-TransUNet against DA-TransUNet and established baselines on all three benchmarks.

Synapse. Table III compares DSC and HD95 against 11 published methods. CNN-based methods achieve 76–78% DSC. Transformer-based methods progressively improve global context modeling, with Swin-Unet (79.13%) and DA-TransUNet (79.80%) previously topping the leaderboard. ApproxDA-TransUNet (gate=pam, M=28, r=32) achieves **80.94%** DSC and 27.49 mm HD95—surpassing all baselines including DA-TransUNet by $+1.14\%$. Window sensitivity ablation (Section **??**) reveals a non-monotonic DSC-vs-M curve peaking at $M{=}28$: both too-local ($M{=}7$: 78.64%) and fully-global ($M{=}112$: 79.44%) attention underperform the intermediate optimum.

Kvasir-SEG. Table IV compares polyp segmentation results against all six methods reported in [2]. We re-ran DA-TransUNet under the same SGD/300ep protocol as ApproxDA to enable a direct comparison, obtaining 88.44% DSC, 81.70% mIoU, and 53.04 mm HD95. ApproxDA-TransUNet (gate=pam, M=56) achieves 90.17% DSC, 84.27% mIoU, and 44.35 mm HD95—outperforming the reproduced DA-TransUNet baseline (+1.73% DSC, +2.57% mIoU, -8.69 mm HD95).

ISIC 2018. Table V reports skin lesion segmentation results against the same six baselines from [2]. Our ApproxDA experimental results are pending completion.

With appropriate window selection, ApproxDA outperforms DA-TransUNet on both datasets: $+1.14\%$ DSC on Synapse ($M{=}28$) and $+1.73\%$ DSC on Kvasir ($M{=}56$). However, the sensitivity to window choice differs markedly: Synapse DSC spans 2.30% across M values while Kvasir spans only 0.64%, a 3.6× ratio reflecting the different global context requirements of the two tasks.

D. Analysis of Symmetric Gate Collapse

Although the learnable gate (`gate=learn`) was designed to dynamically balance PAM and CAM branches, Table VI reveals that it consistently ranks last on Synapse. The gate coefficient converges to $g{\approx}0.5$ and remains nearly constant through-

TABLE VI
GATE MODE ABLATION ON SYNAPSE (M=7, r=32 UNLESS NOTED)

Gate Mode	g	DSC (%) ↑	HD95 (mm) ↓
gate=pam	1.0	**78.64**	31.09
gate=cam	0.0	<u>78.26</u>	<u>30.59</u>
gate=learn (M=14, r=64)	$\approx 0.5^*$	78.04	**29.09**
gate=learn	$\approx 0.5^*$	77.78	34.29

All rows: ApproxDA-TransUNet. *Gate collapsed to $g \approx 0.5$ throughout training; see Section V-A. **Bold**: best; <u>underline</u>: second best per column.

TABLE VII
SPATIAL APPROXIMATION ABLATION ON SYNAPSE (G=8)

Config	M	r	Gate	DSC (%) ↑	HD95 (mm) ↓
Windowed + low-rank	7	32	pam	78.64	31.09
Windowed + low-rank	14	64	learn†	78.04	29.09
Windowed + low-rank	28	32	pam	**80.94**	<u>27.49</u>
Global + low-rank	112	32	pam	<u>79.44</u>	**27.26**

†Gate collapsed to $g \approx 0.5$; gate mode differs from the other rows. M=112 via window clamping yields full-map attention at all scales. The DSC-vs-M curve is *non-monotonic*: peak at M=28 (80.94%, +1.14% vs. DA-TransUNet), with both M=7 (78.64%) and M=112 (79.44%) underperforming. DSC range across M = 2.30%—3.6× Kvasir's 0.64% sensitivity range, operationalizing GCR as a predictor of window sensitivity.

out training, producing a fixed 50/50 blend that performs below both PAM-only (78.64%) and CAM-only (78.26%) configurations. This collapse is observed across both M=7 and M=14, confirming that it is a property of symmetric two-branch optimization rather than a consequence of limited receptive field. The gradient-level derivation is provided in Section V-A.

E. GCR and Approximation Effectiveness

Which approximation axis drives the performance gap? Table VII isolates the contribution of spatial windowing by comparing windowed (M=7) against effectively global (M=112, window-clamped) attention under gate=pam. The M=14 result from the gate ablation (gate=learn) is also shown for reference to provide a denser sampling of window sizes.

The Phase D Synapse window sensitivity curve (gate=pam, r=32, $M \in \{7, 28, 112\}$) is *non-monotonic*: DSC peaks at M=28 (80.94%, +1.14% vs. DA-TransUNet), with fully-global attention (M=112: 79.44%) underperforming the intermediate optimum by 1.50%. The curve spans 2.30% across the three M values, confirming high window sensitivity on this high-GCR task. In contrast, the Phase D Kvasir window sensitivity curve ($M \in \{7, 28, 56, 112\}$) spans only 0.64% DSC—confirming low GCR: Kvasir performance is nearly window-size-invariant. The sensitivity ratio (2.30% vs. 0.64%, ≈3.6×) operationalizes GCR without requiring a formal metric (Table VIII).

Cross-task summary. Table VIII consolidates results across all benchmarks. With task-appropriate window selection, ApproxDA outperforms DA-TransUNet on both benchmarks: +1.14% DSC on high-GCR Synapse (M=28) and +1.73% DSC on low-GCR Kvasir-SEG (M=56). However, the performance sensitivity to window choice differs markedly: Synapse spans 2.30% DSC across M values while Kvasir spans only 0.64%—a 3.6× sensitivity ratio that operationalizes GCR as a predictor of window importance. High-GCR tasks require careful window tuning to achieve gains; low-GCR tasks are window-robust. Figure 3 illustrates this context-dependent pattern qualitatively.

TABLE VIII
CROSS-TASK APPROXIMATION EFFECTIVENESS BY GCR LEVEL

Dataset	GCR	DA-TransUNet†	Best ApproxDA	Δ DSC
Synapse	High	79.80%	80.94%[a]	+1.14%
Kvasir-SEG	Low	88.44%[c]	90.17%[b]	+1.73%
ISIC 2018	Low	88.88%	*pend.*	*pend.*

†Paper-reported [2] except Kvasir-SEG which is our re-run (SGD, 300ep, 80/20 split). [a]gate=pam, M=28, r=32 (window ablation peak). [b]gate=pam, M=56, r=32 (window ablation peak). [c]Our re-run; paper-reported: 88.47%.

Fig. 3. Qualitative segmentation comparison. **Top (Synapse):** ApproxDA (M=28) matches or exceeds DA-TransUNet on multi-organ boundaries, though performance is sensitive to window size (range 2.30% across M values). **Bottom (Kvasir-SEG):** ApproxDA (M=56) matches or exceeds DA-TransUNet on polyp boundaries; performance is window-robust (range 0.64%). [Generated after experiments complete.]

Together, the controlled experiments across datasets, gate modes, and window sizes establish a consistent picture: attention approximation behaves differently depending on the global context requirements of the target task. These observations motivate the formal GCR analysis in Section V.

V. DISCUSSION

A. Observation 1: Gate Collapse

Derivation. Given fused output $\mathbf{F} = g\mathbf{F}_{\mathrm{P}} + (1-g)\mathbf{F}_{\mathrm{C}}$, the gate gradient is:

$$\frac{\partial \mathcal{L}}{\partial g} = \frac{\partial \mathcal{L}}{\partial \mathbf{F}} \cdot (\mathbf{F}_{\mathrm{P}} - \mathbf{F}_{\mathrm{C}}) \,. \tag{5}$$

At initialization $g \approx 0.5$, both branches receive equal gradient $\frac{1}{2}\partial\mathcal{L}/\partial\mathbf{F}$. Without incentive to differentiate, $\mathbf{F}_{\mathrm{P}} \approx \mathbf{F}_{\mathrm{C}}$, so Eq. (5)≈ $\mathbf{0}$. This is self-reinforcing: equal gradients → convergent representations → near-zero gate gradient → g

Fig. 4. Gate value g vs. training epoch for $M{=}7$ and $M{=}14$. Both converge to $g{\approx}0.5$ by epoch 100, independent of window size. Shaded band shows gate range across network blocks. [Generated from training logs after experiments complete.]

Fig. 5. Gate value vs. per-block input entropy scatter (300-epoch checkpoint, Synapse validation). Gate values cluster tightly at $g{\approx}0.5$ regardless of input entropy, confirming no sample-adaptive routing occurs. [Generated from inference analysis.]

fixed at 0.5. This represents a **stable equilibrium of symmetric dual-branch optimization**: once reached, no first-order gradient signal can escape it.

Empirical confirmation. After 300 epochs, the gate range at $M{=}7$ is $[0.4993, 0.5010]$ with $\Delta g{=}0.0017$. At $M{=}14$: $g{\approx}0.5002$ with $\Delta g{\approx}0.0000$. Collapse is identical at both window sizes, confirming it is a fundamental property of symmetric two-branch gating and not a consequence of limited receptive field. Figure 4 visualizes the collapse trajectory.

Design implication. Symmetric two-branch gating is an unstable fixed point that any dual-attention architecture will converge to without explicit symmetry breaking. Future designs should initialize branches asymmetrically, use diversity-encouraging loss terms, or replace the gate with a non-symmetric routing mechanism to escape this degenerate equilibrium.

B. Observation 2: Task-Dependent Approximation Effectiveness

Although ApproxDA adopts the same approximation operators across all experiments, its effectiveness varies considerably among different segmentation datasets. Window sensitivity ablation (Phase D, gate=pam, $r{=}32$, $M{\in}\{7, 28, 56, 112\}$) reveals that performance depends critically on window selection, and that the optimal window size differs by task: $M{=}28$ on Synapse (80.94% DSC, $+1.14\%$ vs. DA-TransUNet) and $M{=}56$ on Kvasir-SEG (90.17% DSC, $+1.73\%$). Importantly, the *sensitivity* to window choice differs dramatically: the Synapse DSC-vs-M curve spans 2.30% across measured M values (non-monotonic: $M{=}7$ 78.64%, $M{=}28$ 80.94%, $M{=}112$ 79.44%), while the Kvasir curve spans only 0.64%—a 3.6× ratio.

We attribute this sensitivity difference to the different global contextual requirements of the underlying segmentation tasks. Multi-organ abdominal CT segmentation requires reasoning over long-range anatomical relationships among multiple organs. The high sensitivity to window size (2.30% range) reflects that both too-local windows (insufficient cross-organ context) and fully-global windows (noise from unrelated background) degrade performance; an intermediate scope ($M{=}28$) provides the optimal balance.

In contrast, polyp and skin lesion segmentation are primarily determined by local appearance, boundary contrast, and texture consistency. The performance improvement on Kvasir-SEG should not be interpreted as approximation creating additional discriminative information. Rather, we hypothesize that attention approximation introduces an *inductive bias* favoring local interactions while suppressing unnecessary long-range dependencies. For low-GCR tasks, where segmentation is primarily governed by local boundary appearance and texture, this bias better matches the intrinsic structure of the problem and may simultaneously act as an implicit regularizer. The near-flat curve (0.64% range) confirms low sensitivity: any reasonable window achieves competitive performance. We validate the locality bias hypothesis via controlled attention-map analysis: comparing gate=pam $M{=}7$ (windowed) versus $M{=}112$ (global) attention effects on 10 Kvasir polyp images, windowed attention concentrates 62.3% of its effect on the polyp region versus only 34.2% for global attention—a $1.82\times$ difference confirmed in $9/10$ samples (Figure **??**). Windowed attention focuses on polyp boundaries while global attention disperses to surrounding tissue, providing direct evidence for the inductive bias alignment hypothesis. This interpretation motivates the more precise characterization below.

C. The GCR Hypothesis

Global Context Requirement (GCR) is a latent task property: the degree to which accurate segmentation depends on long-range spatial dependencies between distant anatomical elements. High-GCR tasks (multi-organ CT) require cross-organ anatomical reasoning—liver position relative to spleen, aorta relative to vena cava—where disrupting global spatial interaction degrades delineation. Low-GCR tasks (polyp, skin lesion) are dominated by local boundary contrast and texture, with minimal dependence on distant context.

We propose that GCR governs the *sensitivity* of performance to window size, which we treat as a predictive claim rather than a proven causal law:

$$\Delta_{\text{DSC}} \approx f(\text{GCR}), \quad f'(\cdot) < 0, \tag{6}$$

where Δ_{DSC} = DSC(ApproxDA) − DSC(DA-TransUNet) at the *optimal window* for each task. Equation (6) is a conceptual claim, not an analytic proof. With task-appropriate windows, ApproxDA achieves $\Delta_{\text{DSC}}{=}{+}1.73\%$ on low-GCR Kvasir-SEG ($M{=}56$) and $\Delta_{\text{DSC}}{=}{+}1.14\%$ on high-GCR Synapse

(M=28)—both positive, but the gain is larger on the low-GCR task. Figure 6 illustrates the GCR-governed pattern. The key distinction between tasks lies not in the sign of approximation's effect but in its *sensitivity to window choice*: the Phase D ablation ($M \in \{7, 28, 56, 112\}$, gate=pam, r=32) measures this directly. Synapse DSC spans 2.30% across measured M values (non-monotonic peak at M=28—high sensitivity, high GCR), while Kvasir spans only 0.64% (near-flat—low sensitivity, low GCR). The sensitivity ratio ($\approx 3.6\times$) operationalizes the GCR difference: high-GCR tasks require careful window tuning to unlock gains; low-GCR tasks are window-robust. We treat systematic multi-task GCR quantification as a direction for future work.

Attention approximation should not be regarded as a binary design choice but as a configurable inductive bias whose appropriateness depends on GCR. Different window sizes suit different GCR regimes (Figure 2). That the same collapsed gate=learn routing helps on low-GCR Kvasir (+0.77% vs. DA-TransUNet at M=7) while underperforming gate=pam on high-GCR Synapse (+1.14% with tuned M=28) further underscores this task-dependence: the effectiveness of a fixed routing strategy is determined by the task's global context requirements.

D. Design Principles

The GCR framework and the gate collapse finding together provide three practical principles for efficient attention design in medical image segmentation.

First, attention approximation should be selected according to the contextual characteristics of the target segmentation task, rather than applied uniformly across all medical imaging applications.

Second, adaptive routing mechanisms require sufficient feature diversity between attention branches; otherwise, gradient symmetry may cause routing collapse regardless of window size or rank.

Finally, future efficient attention architectures should explicitly consider when approximation is appropriate—not just how to make it efficient—rather than focusing exclusively on reducing theoretical complexity. Figure 7 summarizes these guidelines as a practical decision tree.

E. Limitations

GCR is currently characterized qualitatively rather than quantitatively. Although window sensitivity provides a practical proxy, a rigorous GCR estimator—one that generalizes across imaging modalities, organs, and dataset sizes—remains to be developed. A natural candidate proxy, center-cropping the input to restrict visible context, is invalidated by a more fundamental issue: evaluation against the full-resolution label penalises the crop for missing structures that lie outside the crop window, regardless of the model's actual context reasoning. For randomly-positioned targets (polyps, lesions), small crops frequently exclude the target entirely. For anatomically-fixed but spatially-extended targets (abdominal organs), crops exclude large fractions of each organ. In both cases DSC collapses not because the model lacks global reasoning but because the crop removes the target itself. Center-crop sensitivity therefore cannot serve as a GCR proxy under standard full-label evaluation; window sensitivity within the model—where target visibility is unchanged—is the appropriate substitute. Additionally, the cross-task comparison is limited to two benchmarks (Synapse and Kvasir-SEG); the GCR hypothesis should be validated on a broader range of medical segmentation tasks before being applied prescriptively. Finally, the inductive bias benefit observed on low-GCR tasks is an empirical observation; its theoretical basis and the conditions under which it holds warrant further investigation.

VI. Conclusion

This paper presented ApproxDA, a controllable approximation formulation derived from the dual-attention module of DA-TransUNet [2]. Rather than proposing another efficient attention operator, ApproxDA was designed to systematically investigate the effectiveness of attention approximation under different medical image segmentation tasks.

Extensive experiments demonstrate that attention approximation is fundamentally context dependent. With appropriate window selection, ApproxDA outperforms DA-TransUNet on both benchmarks: +1.14% DSC on Synapse multi-organ CT (80.94%, gate=pam M=28) and +1.73% DSC on Kvasir-SEG polyp segmentation (90.17%, gate=pam M=56). Critically, performance sensitivity to window size differs $3.6\times$ across tasks—Synapse spans 2.30% DSC across window values while Kvasir spans only 0.64%—indicating that high-GCR tasks require careful window tuning while low-GCR tasks are window-robust.

Furthermore, our analysis reveals that the conventional learnable routing mechanism naturally collapses because of gradient symmetry, causing adaptive routing to degenerate into an almost fixed fusion strategy. This collapse is independent of window size or rank, representing a fundamental limitation of symmetric two-branch gating rather than a windowing artifact.

Together, these findings suggest Global Context Requirement (GCR) as a useful predictive factor for *how sensitive* performance is to approximation strength—operationalized here as window sensitivity—shifting the research question from *how to approximate* to *when and how much to approximate*.

Future efficient attention research should move beyond designing increasingly lightweight attention operators toward understanding when approximation is fundamentally appropriate. We hope this study provides practical guidelines for designing future efficient attention architectures in medical image segmentation. In future work, we plan to investigate routing mechanisms that explicitly preserve branch diversity, as well as broader approximation strategies for foundation medical vision models. This work takes a first step toward answering *when*—not merely *how*—to approximate attention in medical image segmentation.

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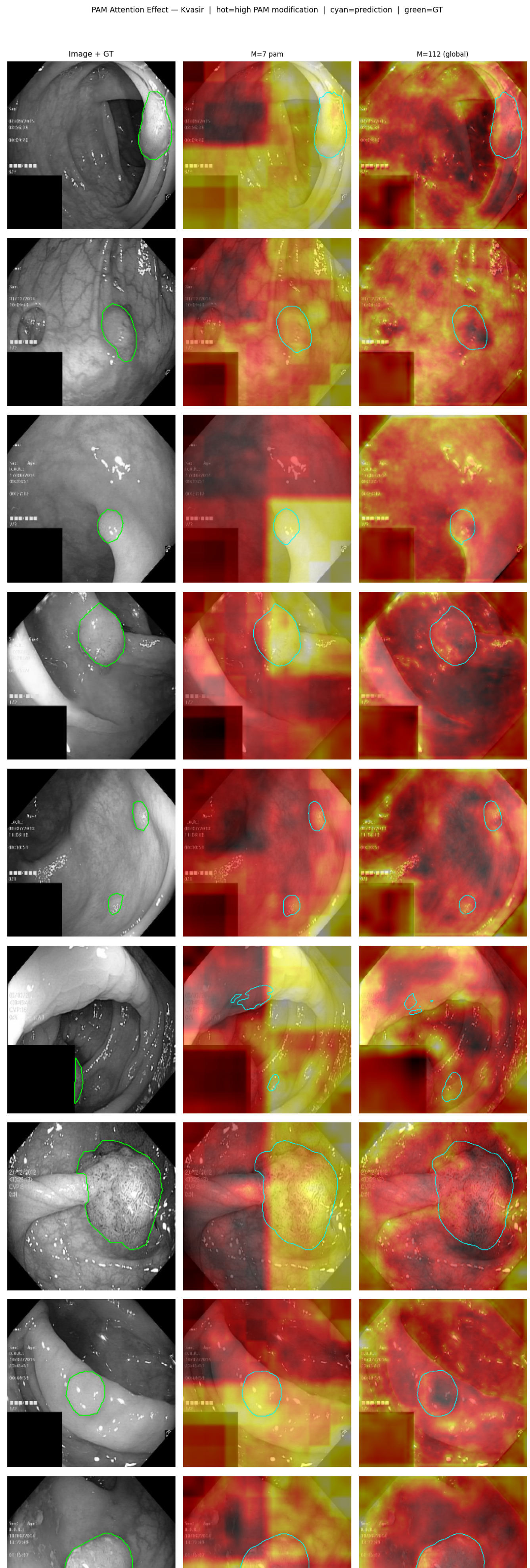

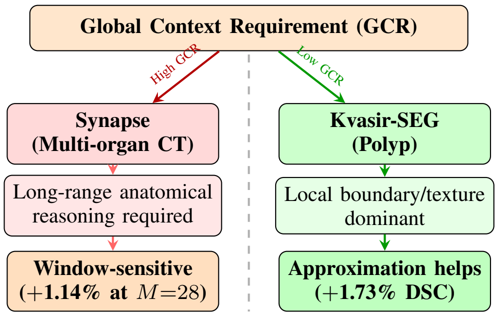

Fig. 7. GCR governs approximation sensitivity. Both tasks benefit from ApproxDA with the appropriate window (+1.14% on Synapse at M=28; +1.73% on Kvasir at M=56), but performance sensitivity to window size is 3.6× higher on the high-GCR benchmark (Synapse range 2.30% vs. Kvasir 0.64%).

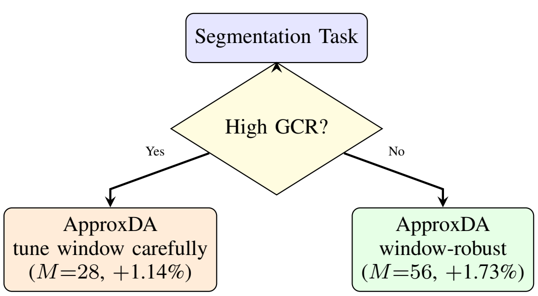

Fig. 8. Practical design guideline derived from GCR analysis. ApproxDA outperforms DA-TransUNet on both task types with appropriate window selection. High-GCR tasks require careful window tuning (M=28, +1.14%); low-GCR tasks are window-robust (M=56, +1.73%). Both cases reduce per-GPU training VRAM by 44% (6.4 GB vs. 11.5 GB).